

Summer 2022 - A	1(c)	i. 50 ms ii. 1200 ms iii. 2270 ms
	2(b)	ii. 40
	2(c)	i. seq: 5550; ack: 10236 ii. seq: 11235; rwnd: 7611
	3(c)	i. 10 ii. 1 iii. 1901 iv. 375
Summer 2022 - B	1(c)	i. 220 ms ii. 1540 ms iii. 4385 ms
	2(c)	i. seq: 5550; ack 10236 ii. seq: 11235; rwnd: 8000
	3(c)	i. 11 ii. 0 iii. 1865 iv. 1992
Fall 2022 - A	1(c)	ii. 5.27 s
	2(c)	i. 76 ms ii. 1645 ms or 1540 ms if 5 ms is not included in RTT calculation iii. 4346 ms
	3(b)	urg flag = 1, urgent pointer = 699
	3(c)	i. seq: 2366; ack = 10266 ii. rwnd = 3020 iii. seq = 2755; ack = 10266

Fall 2022 - B	1(d)	1267.6 ms
	2(c)	i. 104 ms ii. 4904 ms or 4784 ms if 4 ms is not included in RTT calculation iii. 11354 ms
	3(c)	i. seq = 11286; ack = 2138 ii. rwnd = 1384 iii. seq = 2138; ack 11286
Spring 2023 - A	2(c)	i. 184 ms ii. 2602 ms (RTT for DNS will be added as we need to fetch the IP address after inserting the url) iii. 1050.3 Mbps
	3(c)	i. seq = 2871; ack = 9861 ii. seq = 11733; ack = 8205 iii. rwnd = 3777
Spring 2023 - B	1(c)	Each file size will be 4 mb (There is a typo in the question) LAN utilization = 15.6% Access Link utilization = 130.1%
	2(c)	i. 88 ms ii. 10056 ms (RTT for DNS will be added as we need to fetch the IP address after inserting the url) iii. 541.3 Mbps
	3(c)	i. seq = 12097; ack = 9425 ii. seq = 10835; ack = 16542 iii. rwnd = 7338
	2(c)	i. 50 ms ii. 1862 ms iii. 4126 ms

Summer 2023 - A	3(b)	Sf = 731 Sn = 770 Starting byte number of window = 731 Last byte number of window = 930
	3(c)	i. seq = 8765; ack = 1026 ii. seq = 1026 iii. ack = 1026
Summer 2023 - B	2(b)	i. 225 ms ii. 6394 ms iii. 9816 ms
	3(b)	Sf = 1300 Sn = 1400 Starting byte number of window = 1300 Last byte number of window = 1799
	3(c)	i. seq = 2231; ack = 5547 ii. ack = 2563 iii. ack = 3018
Fall 2024 - A	6	i. 2275 ms ii. 17500 ms
	7	i. 105 ms ii. 30 ms
	8	ii. seq = 2332; ack = 2013 iii. rwnd = 11468
Fall 2024 - B	6	i. 294 ms ii. 41142.86 ms
	7	i. 155 ms ii. 35 ms
	8	ii. seq = 1887; ack = 2652 iii. rwnd = 9680

Semester - Set	Question	Answers
Summer 2022 - A	1(a)	i. 19.64.0.0/10 ii. 255.192.0.0 iii. 4194302
	1(b)	Network Addresses without prefix mask [In exams you have to write the prefix mask] 19.64.0.0 19.64.4.0 19.64.6.0 19.64.6.8 19.64.6.12
	4(a)	i. For R1, ip route 0.0.0.0 0.0.0.0 s3/0 or ip route 0.0.0.0 0.0.0.0 192.168.10.2 or ip route 0.0.0.0 0.0.0.0 s1/0 or ip route 0.0.0.0 0.0.0.0 10.10.10.2 or ip route 0.0.0.0 0.0.0.0 s2/0 or ip route 0.0.0.0 0.0.0.0 10.10.20.2
		For R2, ip route 0.0.0.0 0.0.0.0 s0/1 or ip route 0.0.0.0 0.0.0.0 s1/0 or ip route 0.0.0.0 0.0.0.0 10.10.10.1 or ip route 0.0.0.0 0.0.0.0 s2/0 or ip route 0.0.0.0 0.0.0.0 10.10.20.1
	5(b)	FF10::AC19:0:1000:E000

Summer 2022 - B	1(a)	i. 19.253.255.255/15 ii. 15 iii. 19.252.0.2/15
	1(b)	Network Addresses without prefix mask [In exams you have to write the prefix mask] 19.252.0.0 19.252.8.0 19.252.12.0 19.252.12.8 19.252.12.12
	4(c)	i. For R1, ip route 167.18.10.0 255.255.255.240 s3/0 or ip route 167.18.10.0 255.255.255.240 192.168.10.2 For R2, ip route 167.18.10.0 255.255.255.240 s1/0 or ip route 167.18.10.0 255.255.255.240 10.10.10.1 or ip route 167.18.10.0 255.255.255.240 s2/0 or ip route 167.18.10.0 255.255.255.240 10.10.20.1
	5(b)	FF10:FF::AC19:0:0:E000
Fall 2022 - A	1(a)	i. 17.50.127.255/17 ii. 17.50.0.2/17
	1(b)	Network Addresses without prefix mask [In exams you have to write the prefix mask] 17.50.0.0 17.50.4.0 17.50.6.0 17.50.6.16 17.50.6.24
	2(b)	i. 1686 bytes ii. MTU = 230 bytes, offset = 130 iii. MF = 1
	4(a)	i. ip route 1.1.0.0 255.255.255.0 S0 5

Fall 2022 - B	1(a)	i. 200.96.0.0/12 ii. 200.111.255.253/12
	1(b)	Network Addresses without prefix mask [In exams you have to write the prefix mask]: 200.96.0.0 200.96.8.0 200.96.9.0 200.96.9.128 200.96.9.136 200.96.9.140
	2(b)	i. 1756 bytes ii. MTU = 268 bytes, offset = 125 iii. MF = 0
	4(a)	i. ip route 0.0.0.0 0.0.0.0 1.1.1.2
Spring 2023 - A	1(a)	i. 109.64.0.0/10 ii. 4194302
	1(b)	Network Addresses without prefix mask [In exams you have to write the prefix mask]: 109.64.0.0 109.64.8.0 109.64.16.0 109.64.16.8
	2(b)	i. 29430 bytes ii. 356 iii. MF = 0
Spring 2023 - B	1(a)	i. 109.0.15.255/21 ii. 2046
	1(b)	Network Addresses without prefix mask [In exams you have to write the prefix mask]: 109.0.8.0 109.0.10.0 109.0.12.0 109.0.12.8
		i. 23965 bytes

	2(b)	i. 23965 bytes ii. 2640 iii. MF = 0
	4(a)	ip route 43.11.192.0 255.255.255.128 s0/0 or ip route 43.11.192.0 255.255.255.128 s0/1
Spring 2024 - A	1	i. 2048; Note: if the question asked for based on the topology, then 8 ii. Network Addresses without prefix mask [In exams you have to write the prefix mask]: 3.255.192.0 3.255.196.0 3.255.200.0 3.255.202.0 3.255.202.8 3.255.202.12 3.255.202.16
	3	i. 8 ii. 195 iii. 1
Spring 2024 - B	1	i. 8192; Note: if the question asked for based on the topology, then 16 ii. Network Addresses without prefix mask [In exams you have to write the prefix mask]: 1.2.128.0 1.2.136.0 1.2.140.0 1.2.144.0 1.2.144.8 1.2.144.12
	3	i. 13 ii. 135 iii. 1

Fall 2024 - A	1	i. 255.255.192.0 ii. 42.1.0.0/18 iii. Network Addresses without prefix mask [In exams you have to write the prefix mask]: 42.1.0.0 42.1.8.0 42.1.12.0 42.1.16.0 42.1.16.4
	3	i. 5 ii. 190 iii. 612
	4	i. ip route 192.168.96.0 255.255.255.0 192.168.10.97 2 ii. ip route 192.168.96.0 255.255.255.0 S4 3
Fall 2024 - B	1	i. 32766 ii. 32.99.127.255/17 iii. Network Addresses without prefix mask [In exams you have to write the prefix mask]: 32.99.0.0 32.99.8.0 32.99.16.0 32.99.16.8 32.99.16.12
	3	i. 4 ii. 1525 iii. 609
	4	i. ip route 0.0.0.0 0.0.0.0 S2 10 or, ip route 0.0.0.0 0.0.0.0 S1 10 ii. ip route 0.0.0.0 0.0.0.0 192.168.12.194 11 or, ip route 0.0.0.0 0.0.0.0 192.168.10.101 11